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Student Submissions — Application Based Questions

SIMATS Engineering • CSE • CSA0802 — Python Programming • 05-08-2026 • Category: Application Based Questions • Student Submissions Report

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Q1. Student Grade Book Application

CODE

```

students = {
    "Aarav": [85, 90, 78, 92, 88],
    "Diya": [95, 92, 89, 91, 94],
    "Kabir": [45, 55, 60, 40, 50],
    "Meera": [70, 65, 72, 68, 75],
    "Rohan": [60, 58, 62, 55, 59]
}

def calculate_grade(average):
    if average ≥ 90:
        return "O"
    elif average ≥ 80:
        return "A+"
    elif average ≥ 70:
        return "A"
    elif average ≥ 60:
        return "B+"
    elif average ≥ 50:
        return "B"
    else:
        return "F"

results = {
    name: {
        "marks": marks,
        "total": sum(marks),
        "average": round(sum(marks) / len(marks), 2),
        "grade": calculate_grade(sum(marks) / len(marks))
    }
    for name, marks in students.items()
}

topper = max(results, key=lambda name: results[name]["average"])

num_subjects = len(next(iter(students.values())))
subject_means = []
for i in range(num_subjects):
    subject_marks = [marks[i] for marks in students.values()]
    subject_mean = sum(subject_marks) / len(subject_marks)
    subject_means.append(subject_mean)

highest_subject_index = subject_means.index(max(subject_means))
subject_names = [f"Subject {i+1}" for i in range(num_subjects)]

sorted_results = sorted(results.items(), key=lambda item: item[1]["average"], reverse=True)

print("=" * 70)
print(f"{'Name':<10}{'Marks':<25}{'Total':<8}{'Average':<10}{'Grade':<6}")
print("=" * 70)

for name, data in sorted_results:
    marks_str = str(data["marks"])
    print(f"{'name':<10}{'marks_str':<25}{'data['total']':<8}{'data['average']':<10}{'data['grade']':<6}")

print("=" * 70)

print(f"\nClass Topper: {topper} with an average of {results[topper]['average']}")
print(f"Subject with highest overall mean: {subject_names[highest_subject_index]} "
      f"(Mean = {round(subject_means[highest_subject_index], 2)})")

print("\nSubject-wise Mean Marks:")

```

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```
for i, mean in enumerate(subject_means):
    print(f" Subject {i+1}: {round(mean, 2)}")
```

OUTPUT (no output)

Q2. Debugging and Rewriting

CODE

```
accounts = {}

def create_account(acc_no, name, balance):
    accounts[acc_no] = {'name': name, 'balance': balance, 'txns': []}

def deposit(acc_no, amount):
    if acc_no in accounts:
        accounts[acc_no]['balance'] += amount
        accounts[acc_no]['txns'].append(('Deposit', amount))

def withdraw(acc_no, amount):
    if acc_no in accounts:
        if accounts[acc_no]['balance'] >= amount:
            accounts[acc_no]['balance'] -= amount
            accounts[acc_no]['txns'].append(('Withdraw', amount))
        else:
            print('Insufficient balance')

def mini_statement(acc_no):
    for t in accounts[acc_no]['txns'][-5:]:
        print(t)

create_account(1001, 'Ravi', 10000)
deposit(1001, 5000)
withdraw(1001, 20000)
withdraw(1001, 3000)
mini_statement(1001)
print('Balance:', accounts[1001]['balance'])
```

OUTPUT (no output)